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Design and Prototyping of a Platform for Academic-Industry Project Collaboration

Software Requirements Specification for a Project Studies Platform

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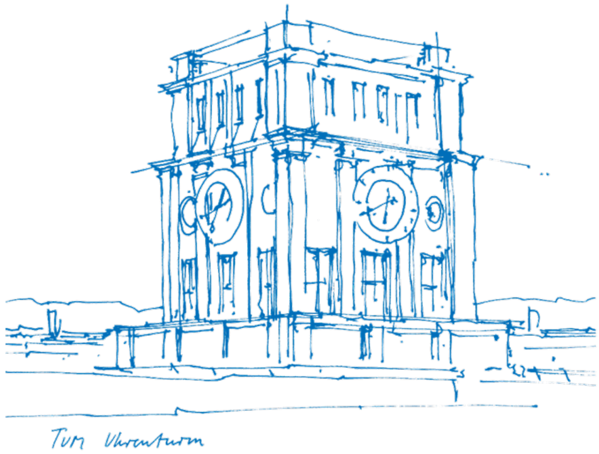

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1. Introduction

1.1. Purpose of this Document

This specification defines the requirements for a project studies platform, developed as part of a bachelor's Project Study at TUM Campus Heilbronn in collaboration with BirdVision. It is adapted from ISO/IEC/IEEE 29148, scaled down and extended with a literature review section to meet the academic requirements of a project study report. Every requirement in Section 6 and Section 7 is traceable back to its source and forward to its prototype coverage in the deployed prototype[1], per Section 9. That traceability, more than any single requirement, is what this document is meant to demonstrate.

1.2. Project Background and Motivation

A project study for TUM School of Management students is a research or practical project, carried out by a student team in collaboration with a company and a supervising professorship. This project, "Project Studies Platform for BirdVision", was assigned to build a platform supporting a project study process similar to this one, which this document itself is a product of.

Companies, professors, and students currently coordinate project studies at TUM Campus Heilbronn in three ways: university staff sourcing companies directly and sending out given topics to professors, professors and companies agreeing between themselves, or students agreeing with a company and finding a supervising professor on their own. These processes are decentralized and realized via a variety of mediums. Recurring pain points arising from the status quo were identified through semi-structured interviews and written responses, which this document translates into a scoped set of requirements.

1.3. Scope of the Platform

The scope of the platform is limited to the project study process at TUM Campus Heilbronn, rather than industry-academia collaborations in general. The reasoning behind this scope and discussion on how this platform can be adapted for different institutions is written out in sections 4.4 and 4.5. Broader findings from the interviews are not excluded and presented in Section 5.

To deliver on the project goal of a testable MVP, this document is accompanied by a non-functional prototype.

1.4. Definitions, Acronyms, and Abbreviations

- **SRS**: Software Requirements Specification
- **FR** / **NFR**: Functional Requirement / Non-Functional Requirement
- **MoSCoW**: Prioritization scheme: Must, Should, Could, Won't (Section 6.1)
- **NDA**: Non-Disclosure Agreement
- **GDPR**: General Data Protection Regulation

[1] https://nan1220.github.io/Collaboration_Platform/

- **Professorship**: Used in this document to refer both to the professor and the PhD candidates/research associates working for the chair; professor/researcher or professor/supervisor is sometimes used instead

1.5. Document Overview

Section 2 reviews the literature behind the project's methodological choices. Section 3 goes over the interview methodology, sample, and its limitations. Section 4 describes the platform's overall scope, boundaries and transferability. Section 5 presents the pain points the requirements are derived from. Section 6 and Section 7 specify the functional and non-functional requirements. Section 8 provides system models. Section 9 traces requirements back to their source and forward to prototype coverage. Section 10 discusses risks and conclusions.

2. Literature review

In this section we review the academic grounding for the project's methodological choices.

2.1. Requirements Engineering Grounding

2.1.1. Selection and Adaptation of the Requirements Engineering Standard

This specification is based on ISO/IEC/IEEE 29148:2018 (Society, 2018) – the current international standard for requirements engineering. 29148 covers the requirements process as a whole, including elicitation, analysis and validation, in comparison to older versions. Consequently, such a form is more appropriate for this project, since much of what follows concerns how requirements were derived from stakeholder interviews/responses and how strongly each is supported.

2.1.2. Interview-Based Requirements Elicitation

Semi-structured interviews were used to explore stakeholder needs without assuming predefined system requirements. Ferrari et al. (2022) show that interviews support an iterative elicitation process in which stakeholder initial ideas are progressively clarified and refined into requirements through the dialogue.

2.1.3. Sample Adequacy and Thematic Saturation

Thematic saturation, the point at which no new codes emerge from additional interviews, is a recurring benchmark in qualitative research according to MacQueen et al. (1998), though what counts as "enough" varies with what is being measured.

Guest et al. (2006) found little new information after 12 of 60 interviews, Hennink et al. (2017) found code saturation at 9 interviews, but 16–24 were needed for a rich understanding of those codes. Hagaman & Wutich (2017), working across four sites, found 16 interviews or fewer sufficient within a single group, but cross-group themes required 20 to 40.

Taken together, this suggests a lower bound around 9 to 12 interviews for basic code saturation within a stakeholder group, and a more defensible target of 16 or more for meaningful within-group understanding, with 20 to 40 needed for cross-group findings to be considered stable.

2.1.4. Requirements Traceability

Requirements traceability is the ability to follow a requirement forward into design and implementation and backward to the source it came from. Gotel & Finkelstein (1994), drawing on interviews and questionnaires with over 100 practitioners, distinguish these two directions and argue that the backward one is where projects usually fail.

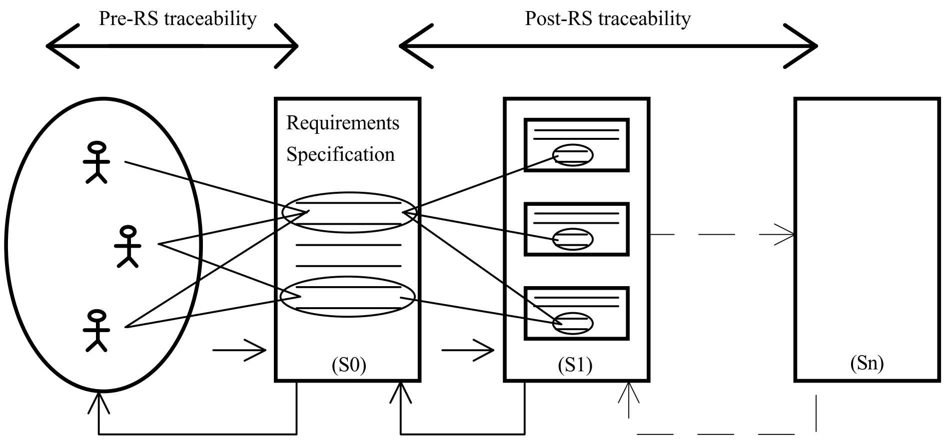

Figure 2.1: Illustration of the two directions of requirements traceability from Gotel & Finkelstein (1994)

Post-specification traceability, linking requirements to what was built from them, is comparatively well supported by tools. Pre-specification traceability, linking a requirement back to the stakeholder statement and rationale that produced it, tends to be neglected, and most of the problems attributed to poor traceability turn out to originate there.

Section 9 is organized along both directions: Section 9.1 provides the pre-specification record, tracing each requirement back to its pain point and interview material, while Section 9.2 provide the post-specification record, mapping requirements forward to prototype screens. Section 9.3 accounts for pain points without requirements and requirements without implementation.

2.1.5. Requirements Prioritization

Prioritization techniques differ mainly in how much they demand of the team using them. Reviewing the field, Achimugu et al. (2014) find that pairwise methods such as the Analytic Hierarchy Process become impractical as requirement counts grow, and that cost-value approaches depend on effort and benefit estimates unavailable at this stage of a project. MoSCoW asks for neither, sorting requirements into four ordinal classes on judgement alone, which suits a small team working to a fixed deadline on requirements that have not been validated. The criteria used for that sorting are set out in Section 6.1.

2.2. Academia-Industry Collaboration

2.2.1. Collaboration friction as an established phenomenon

Collaboration between universities and companies has been studied extensively, and the difficulties it produces are documented rather than incidental. Ankrah & AL-Tabbaa (2015) synthesise a fragmented body of work into a process framework covering why the two sides collaborate, the forms collaboration takes, and the organisational obstacles that recur across national contexts. Coordination and communication problems appear throughout that literature, independent of country, discipline and collaboration format. Since this problem is documented across many institutions, the difficulties reported in Section 5 are treated as a specific case of it rather than as a local peculiarity.

2.2.2. Barriers and information asymmetry between stakeholder groups

Bruneel et al. (2010) separate two kinds of barrier to university-industry collaboration. Orientation-related barriers arise from differences in what each side is trying to achieve and on what timescale. Transaction-related barriers arise from the mechanics of working together, including administration and intellectual property. Prior collaboration experience reduces the first kind, while trust between the parties reduces both. Difficulties such as identifying the right contact person or establishing what expertise is available fall into the second category, which is where a platform can plausibly intervene.

2.2.3. Divergent motivations of academic and industry partners

The two sides enter collaboration for different reasons. Perkmann et al. (2013) show that academic engagement is driven largely by research-related motives, including access to data, funding and problems worth studying, while firms pursue capability, talent and commercial application. Neither set of motives is illegitimate, but they are not automatically aligned, and the semester rhythm that governs university work rarely matches company project timelines. A platform mediating between the two therefore cannot assume a shared objective; it has to make each side's constraints legible to the other. Section 5.4.3 returns to this point where the mismatch surfaced in the interview data.

2.2.4. Platforms and intermediaries as a category of solution

Where two parties struggle to find and evaluate each other, a third party can lower the cost of doing so. Howells (2006) develops a typology of innovation intermediaries and defines them as bodies acting as brokers between parties in the innovation process, performing functions such as scanning, matchmaking and coordination. The platform specified in this document sits in that category, with the difference that it serves three groups rather than two.

3. Research Approach and Limitations

3.1. Interview Methodology

The study followed a sequential design. The requirements engineering literature reviewed in Section 2.1 informed the interview guide, and a scoping discussion with BirdVision fixed what the client needed from the study. Stakeholder groups were then defined, participants recruited and interviews conducted. Responses were consolidated into a single matrix, analysed by group, and translated into requirements, from which the prototype, the traceability table, the use case diagrams and the key user flows were produced.

Interviews were semi-structured, using a common guide across all groups so that responses stayed comparable, with follow-up questions free to depart from it wherever a participant raised something unanticipated. Some responses were collected via email when the participant was not available for an interview, resulting limitations are discussed in the following section. The guide covered four areas:

1. **Current collaboration experience** — what forms of collaboration the participant had taken part in, and how they found them.
2. **Motivation** — what makes collaboration worth entering, and what makes a project attractive.
3. **Problems and barriers** — what goes wrong during collaboration, and what prevents people from engaging at all.
4. **Platform and solution needs** — asked last, so that earlier answers were not framed around a software solution.

The first three guiding questions explore how collaboration currently works and why it is difficult, while the last one concerns the platform directly.

Four stakeholder groups were distinguished: students, companies, professorships and university staff. Staff were separated from professors because their role is administrative rather than academic, and their experience and motivations proved to differ accordingly. Participants were recruited through campus contacts or via publicly available institutional email addresses. Additionally, most company representatives were first contacted at TUM Career Factory. Section 3.2 discusses the resulting composition and limitations.

The response matrix had participants as columns and the four areas as rows, allowing each theme to be read across all participants and across groups. Section 3.3 describes how the requirements were derived from it.

3.2. Sample Composition and Limitations

Twenty-six semi-structured interviews and written responses were gathered across four stakeholder groups. Table A.1 lists all participants; the distribution is summarized below.

Table 3.1: Interview sample by stakeholder group

Group	n	Institutional background
Students	13	9 TUM Heilbronn, 2 TUM Garching, 1 RWTH Aachen, 1 Reutlingen
Companies	5	Startup, industrial manufacturing, software, energy management, IT services
Professorships	6	2 TUM Heilbronn, 1 TUM Munich, 1 RWTH Aachen, 1 Koblenz, 1 Passau
University staff	2	2 TUM Heilbronn

The sample spans five universities and seven campuses. Eight of the 21 university-affiliated participants are based outside TUM Campus Heilbronn, so where the same difficulties recur across institutions, they are unlikely to be local artefacts.

Some responses were collected in writing over email rather than through a live interview, where a participant was not available for a call. This affects the core advantage of the semi-structured format described in Section 2.1.2: a written response cannot be followed up on the way a live interview can, so points raised only briefly could not be probed further, and answers that departed from the guide's four areas were not clarified in real time. Written responses are otherwise treated the same as oral ones throughout the analysis, but should be read with this constraint in mind.

Two limitations shape how the company findings should be read. Most company representatives were recruited at Career Factory, a university-company event on the Bildungscampus, so the sample includes no firms that have never collaborated with a university or have stopped doing so. More importantly, four of the five hold roles in human resources, apprenticeship management or campus relations. Their interest in the university is oriented towards recruiting, and the requirements derived from this group reflect that. Needs around research cooperation, confidentiality and intellectual property are under-represented. The founder interviewed as C1 offers a partial counterweight, having neither a recruiting function nor an employer-branding budget, and is treated in Section 6 as a contrasting case rather than as one observation among five.

The company, professor and staff groups are small, and findings specific to them are treated as indicative. The staff group is smallest at two, though both are the administrators who currently run the project-study process at TUM Campus Heilbronn, so the group covers the relevant population rather than a sample of it.

This background of existing engagement is unevenly distributed across the sample. Several students and professors interviewed had no established route into university–industry collaboration, so their accounts include the perspective of those outside it. All five company representatives, by contrast, were already engaged with universities at the time of the interview. The company data therefore describes how engagement works for firms that have already committed to it, and says little about what would bring in a firm that has not.

3.3. From Interview Insights to Requirements

The analysis started from the response matrix described above. Each theme was read across all 26 participants, first within a stakeholder group and then across groups, and recurring statements were grouped into pain points. Pain points named by only one group are reported under that group in Section 5, while those raised by more than one group are reported once among the cross-cutting findings at the end of that section. Each pain point lists the participants who raised it, so a reader can see how many independent sources stand behind it.

Requirements were then derived from these pain points. The two sets do not match one to one. Some pain points lie outside what a platform can fix, and those are accounted for in Section 9. Some requirements, in turn, come from elsewhere. Each requirement therefore states what kind of evidence it rests on:

Table 3.2: Evidence basis recorded for each requirement

Basis	Meaning
Interview	Taken from one or more interview statements, cited by participant ID.
Process documentation	Taken from how the project-study process currently works at TUM Campus Heilbronn, rather than from interview data.
Team decision	A choice made by the authors, where neither source supplied a requirement. Recorded as Team Decision in the traceability table (Table A.3), with the reasoning given at the requirement itself.

Two requirements rest on a team decision rather than on interview data or the current process: Student Profiles (FR-9) and Dual Approval (FR-15). Each is argued for where it is specified, and FR-15, which proposes a change to how student selection currently works, is discussed again in Section 10.

Another attribute is recorded in order to distinguish transferability between institutions: whether the requirement depends on TUM's administrative arrangements or would apply at any university.

The prototype was built from the requirement set as a whole rather than requirement by requirement. The mapping from requirement to screen in Section 9 was therefore drawn up after implementation, not before it.

Two limits follow. The requirements were never taken back to participants for confirmation, and the step from interview statement to pain point rests on the authors' reading of the material. Together with the sample limitations described above, this is why the requirements in Section 6 and Section 7 are presented as informed proposals. Each one can be traced to its evidence, and the strength of that evidence is recorded, but none has been checked against the people it came from.

4. Overall Description

4.1. Product Perspective

The platform is a new, self-contained system rather than an extension of existing university software. It acts as an intermediary among parties who currently find one another through scattered institute pages, personal contacts, job boards and events, and its purpose is to reduce the cost of that search and standardise the information exchanged once contact is made.

What exists at the end of this project study is a prototype. It demonstrates the interfaces and flows specified here, but implements no persistent storage and no integration with existing institutional processes.

4.2. Stakeholder Groups, Roles and Motivations

- **Students:** search for projects, apply and carry out the work, motivated by practical experience, industry contact and, in several cases, employment prospects. Their engagement is conditional — several said they would withdraw from a project seen as serving only a commercial interest.
- **Companies:** post projects and receive results, motivated by talent contact, access to expertise and usable outcomes. The companies interviewed were oriented mainly towards recruiting, and the requirements derived from them reflect that.
- **Professors and academic researchers:** supervise and assess projects, motivated by access to real problems and by teaching value. Their constraint is time.
- **University staff:** administer the process, matching requests to supervisors and tracking progress. Their main obstacle is obtaining timely responses from professorships.

These motivations are not aligned by default. The platform does not assume they are; its function is to make each group's constraints visible to the others.

4.3. Assumptions and Constraints

The specification assumes a university runs the platform, not a commercial third party. This matters because several requirements depend on the operator being able to confirm that a profile really does belong to a member of the university, and only the university itself holds that information. We further assume that every user has an affiliation which can be verified, and that each project study runs to a timeline agreed between the supervising professor and the company rather than to a fixed institutional calendar.

There is one more assumption, about how selection currently works. Students at the TUM Global Center for Family Enterprise are chosen by the university alone, with no involvement from the companies whose projects they apply to. Section 6 contains a requirement that would change this. We have flagged it there as a proposed change to the process rather than presenting it as something the platform merely automates.

Our constraints were those of a single-semester project study, with no budget and a small team. Everything here rests on interview data. We could not consult institutional process documentation beyond what participants described to us, and nobody with legal expertise reviewed the data-protection and confidentiality questions that operating such a platform would raise.

4.4. Scope Boundary: TUM-Specific Focus

The platform is specified against one process in particular: the project study process at TUM Campus Heilbronn. Administrative arrangements vary a great deal between universities, and a specification broad enough to fit all of them would be too vague to build from. We chose depth against a process we could examine closely.

The problems it addresses are another matter. We modeled a local workflow, but the difficulties it responds to came up at every university in our sample.

4.5. Transferability to Other Institutions

Each requirement is tagged according to how far it depends on local arrangements:

Table 4.1: Transferability tags applied to requirements

Tag	Meaning
Generalizable	Would apply at any university.
Adaptable	Depends on an arrangement other institutions also have in some form; needs reconfiguring rather than redesigning.
Institution-specific	Tied to how TUM Campus Heilbronn organises the process; would require rework elsewhere.

These tags feed the prioritization scheme in Section 6.1: requirements that are generalizable and grounded in cross-cutting pain points rank above those that are institution-specific or supported by a single group.

4.6. Out-of-Scope Items

Outside the scope of this specification and the prototype: persistent storage and backend services, integration with university administration systems, and how the platform would acquire its first users on each of the three sides.

5. Stakeholder Needs

Four stakeholder groups were tracked during recruitment: students, companies, professorships, and university staff, reflecting their distinct administrative and academic roles. For reporting pain points in Section 5, however, staff are grouped with professors rather than presented separately, given the small size of the staff sample and the TUM Campus Heilbronn-specific nature of their perspective.

Each pain point in this section is presented with the interview ID it was sourced from. See Table A.1 for anonymized interview counts per stakeholder group and matching interview IDs.

The findings are sourced in the context of industry-academia collaborations in general, thus might not be entirely relevant for project studies specifically. These findings are nonetheless documented in full below.

5.1. Student Pain Points

5.1.1. Discoverability

Students consistently report that there is no clear, centralized listing for company projects. Project opportunities are described as scattered across fragmented institute websites or buried in static job boards (S6, S7, S8).

5.1.2. Missing or insufficient listing information

Project information about workload, technical requirements, deliverables, and whether the company genuinely prioritizes the project's outcome is frequently unavailable before applying (S8, S9). Company name, expected outcomes, and required skills were independently named as the minimum information needed to make an informed decision (S7, S9).

5.1.3. Communication and responsiveness

Slow communication loops and vague project scope were named as reasons a student would abandon an application outright (S8). Separately, students report response delays during the application process and a lack of feedback following rejection (S12, S13).

5.1.4. Language barrier

German-language requirements were named as a disadvantage for international students on multiple, independent occasions (S5, S7, S10).

5.1.5. Team formation

A student named the inability to find a team as something that would stop them from applying altogether, and separately requested an automated team-matching feature (S10).

5.1.6. Application process friction

One student described having to repeatedly re-enter the same information across each company's own application platform, alongside generic, under-informed initial screening and a lack of feedback after rejection (S1). A separate student reported that project details were withheld pre-NDA, leading to a fundamentally different understanding of the task once disclosed (S9). Yet another student described the challenges they faced in finding an academic supervisor for their company project (S12).

5.1.7. Value and goal alignment

Several students indicated they would disengage from a project perceived as serving only the company's commercial interest without genuine educational or research value (S6, S8). This pattern recurs across other stakeholder groups and is developed further in Section 5.4.

5.2. Professor/University Pain Points

5.2.1. Information standardization

Professors report that companies frequently fail to provide sufficiently detailed project briefs, desired area of expertise, objective, and data availability, leaving task scope unclear for both supervisor and student (P3, P4, P6).

5.2.2. Contact and coordination burden

Locating appropriate contacts and coordinating a collaboration was named as time-consuming and requiring substantial advance planning (P1). On the administrative side, obtaining a timely response from professorships regarding supervision requests was named as the primary obstacle encountered by university staff coordinating intake (U1, U2).

5.2.3. Students' engagement with supervisors during the project

One professor observed that students frequently under-communicate with their academic supervisor once a company relationship is underway, directly linking this to reduced outcome quality, and has introduced mandatory monthly check-ins as an informal mitigation (P4).

5.2.4. Outcome measurement

Two different professors reported that there is no formal mechanism for assessing whether the effort invested in a project study was proportionate to its value. One describes this as entirely unmeasured, the other as narrative rather than KPI-based (P1, P5). This pattern recurs in Section 5.4.

5.2.5. Resource pressure

Time and financial resources dedicated to these collaborations were named as the first budget item companies tend to cut, constraining the professor's ability to sustain partnerships (P5).

5.3. Company Pain Points

5.3.1. Contact identification and structural complexity

Identifying the correct contact person and aligning expectations, timelines, and available resources was the most frequently cited difficulty (C3, C4). University structures were separately described as complex and inconsistent between institutions (C4), directly reinforcing the transferability discussion in Section 4.4-4.5.

5.3.2. Missing information when searching for partners

Companies report a lack of visibility into contact details, areas of expertise, ongoing projects, and collaboration opportunities (C4), as well as the absence of a clear overview of available posting/collaboration options (C5).

5.3.3. Output quality and delivery friction

For one company, the central concern was not access to universities but the practical usability of delivered outcomes, compounded by software-tooling mismatches between student and company environments (C2). A second company reported an inability to assess student or study quality in advance, which discourages deeper engagement. (C5).

5.3.4. Need for trust and verification

Companies want assurance that student and professor profiles are genuinely affiliated with their stated institution, citing existing third-party verification services elsewhere as a reasonable precedent (C4).

5.3.5. Screening signal degradation

One company noted that student CVs are increasingly perceived as AI-generated and therefore less reliable as a screening tool, expressing a preference for more direct channels, including contact with student clubs, over formal university structures (C3).

5.4. Cross-Cutting Findings (pain points recurring across all three groups)

5.4.1. Contact-person and reachability breakdown

Identifying and reaching the correct person was named independently by professors (P1), companies (C3, C4), and university staff (U2, U3). It is the most broadly and heavily evidenced finding in the dataset, spanning three stakeholder groups.

5.4.2. Bidirectional information insufficiency

Every stakeholder group reports the same underlying gap directed at a different counterpart: students want more complete project information from companies and professors (S7, S9); professors want companies to provide more structured project detail (P3); companies want more visibility into student and professor profiles, expertise, and ongoing activity (C4). The specific direction differs but the pattern recurs across all three groups.

5.4.3. Academia-industry goal misalignment

This is the most substantively supported cross-cutting finding. The university research staff perspective names it directly (U1), one company frames it as their central complaint about output usability (C2), a second names it explicitly as a reason to disengage (C4), and it recurs independently across multiple students, either as a disengagement trigger (S6) or as lived experience (S8, S11).

5.4.4. Outcome quality

Two professors report no formal mechanism for evaluating whether a project study delivered proportionate value (P1, P5); this connects directly to one company's independent complaint that project studies deliver limited usable insight (C2) and another professor's complaint that outcome quality is highly dependent on student engagement (P4).

5.4.5. Slow or unresponsive communication

Named as an application-ending concern by students (S8, S12, S13) and, from the opposite side of the same process, as staff's core administrative obstacle in obtaining professor responses (U2, U3).

6. Functional Requirements

6.1. Prioritization Scheme

Requirements are prioritized using MoSCoW (Must, Should, Could, Won't), a judgement-based scheme, which is suited to the context of this project. Priority is assigned according to two criteria, applied in the following order: 1) Structural necessity. A requirement is Must-priority if the core application-to-supervision workflow cannot operate from start to finish without it. This holds true no matter how many interviews support it directly. 2) Strength of evidence. Among the requirements that are not structurally necessary, those founded on a cross-cutting finding or supported across multiple interviews are ranked as Should. Requirements based on a single citation, a team decision, or those that meet a narrower need are ranked as Could. Requirements that are clearly out of scope are marked as Won't in Section 4.6.

Transferability (Section 4.5) is noted for each requirement, but it does not influence this prioritization. The two attributes address different questions, and a requirement can do well in one without doing well in the other.

6.2. Access Features

6.2.1. FR-1, Role-Based Access (Must)

There are four different login/user types: Student, Company, Professor, and Staff. Each type has a unique dashboard. Although no interview explicitly requested this, every stakeholder-specific feature depends on it. Without separate roles, the platform

cannot display different information to different users. This makes it a necessary requirement rather than an optional addition.
Source: Process Documentation (design decision, structural)

6.3. Company Features

6.3.1. FR-2, Company Project Submission Portal (Must)

The following fields are required: Project Title, Required Area of Expertise, Project Background and Objective, Project Deliverable, Available Company Resources, Required Student Skills, Group Size, Company Contact Person. This is adapted from the current project proposal sheet.
Source: S7, S8, S9, P3, P4, P6

6.3.2. FR-3, Submission Review and Approval (Should)

New company project submissions are marked as ‘Pending’. Once approved by staff, their status changes to ‘Approved’, and they become visible to professors for a supervision decision.
Source: Process Documentation

6.3.3. FR-4, Company Browse Student Topics (Must)

Companies can browse topics submitted by students. Acceptance requires the company to fill in mandatory fields (company name, company contact information), and the completed entry becomes visible to professors.
Source: C2 (directly suggested this feature), C5, S12

6.4. Professor Features

6.4.1. FR-5, Professor Profiles and Expertise Matching (Should)

Professor profiles will display their chair and area of expertise, sourced from institution logins. When logged in, professors will see project entries matched to their expertise.
Source: P3, P4, U1, U2

6.4.2. FR-6, Professor Supervision Take-on (Must)

To take on a project — either submitted by a company (FR-2) or a student topic accepted by a company (FR-4) — the professor provides chair contact information, application deadlines, and required documents. Once the professor takes on supervision, the project’s status changes to ‘Open’ and it becomes visible to students for application.
Source: Process Documentation

6.4.3. FR-7, Direct Project Submission (Should)

When a professor and a company agree directly on a project study, the professor will submit it with the required fields filled in for students to apply.
Source: Process Documentation

6.5. Student Features

6.5.1. FR-8, Student Project Submission Portal (Must)

The mandatory fields include: Project Title, Required Area of expertise, Project Background and Objective, Project Deliverable, Required Company Resources, Group size, Student Team Contact Person (adapted from the current project proposal sheet).
Source: C2, C4, C5, S12

6.5.2. FR-9, Student Profiles (Should)

Student profiles will list their program or degree, sourced from institution logins. This will help match students with teams and allow professors and companies to filter students by their program.
Source: Team Decision

6.5.3. FR-10, Student Team Matching (Could)

A student can set their profile status to 'seeking teammates' and include a short message and contact information. Profiles can be filtered by this status.
Source: S10

6.5.4. FR-11, Student Browse Available Projects and Apply (Must)

Students can view available projects with complete information, including company details, expertise required, application deadlines, skills needed, and contact person. They can also apply for a given entry by submitting the required documents.
Source: S6, S7, S8, S9

6.5.5. FR-12, Application Status Visibility (Should)

Students will see a simple accepted/rejected status for each application.
Source: S1, S8, S12, S13

6.6. General Features

6.6.1. FR-13, Status-Change Notifications (Should)

Automated notifications, such as emails, will be sent when an application's status changes or when a listing closes.
Source: S8, U1, U2

6.6.2. FR-14, Lightweight Progress Check-in (Could)

A simple check-in mechanism will allow interaction between students and supervisors during the project.
Source: P4

6.6.3. FR-15, Dual Approval (Professor + Company) (Could)

A student's application will require approval from both the professor and the company before it is finalized. At TUM Campus Heilbronn, the selection of students is currently handled solely by the university, with no formal role for companies in reviewing individual applicants. This proposed change aims to give companies input on who joins their projects, which could lead to higher satisfaction with the outcomes. However, this change may slow down the process and add a coordination step between professors and companies, which are recognized costs of the benefit.
Source: Team Decision

6.6.4. FR-16, Manual Multi-Offer Decision (Should)

If a student is accepted to multiple projects, they will manually confirm one and withdraw from the others. This follows from FR-12 and FR-13: once the application statuses are real, a student accepting two projects simultaneously is a situation the platform must manage.
Source: Process Documentation

6.7. Administrative Staff Features

6.7.1. FR-17, Staff Master Dashboard (Should)

Staff will be able to view all project studies across every status: Pending, Approved, Open, Ongoing, and Complete.

- **Pending** projects have been submitted by companies but have not yet been approved by staff.
- **Approved** projects have passed staff review and are visible to professors for a supervision decision.
- **Open** projects have been taken on by a supervisor and are available for students to view and apply to.
- **Ongoing** projects have a student team currently working on them.
- **Complete** projects have been completed and the final deliverable submitted.

Source: U1, U2

6.7.2. FR-18, Filter Incoming Submissions (Should)

Staff will have the ability to filter and search incoming company submissions during reviews. The volume of submissions will scale with FR-3. Filtering will ensure that manual reviews do not become bottlenecks.
Source: Process Documentation

7. Non-Functional Requirements

7.1. Usability

Four groups use the platform: students, companies, professors, and university staff. Not all of them work with software every day, professors and administrative staff least of all. Each role sees only the functions it needs (FR-1, Role-Based Access), so no one has to learn the parts of the system that belong to someone else. Every listing shows the same fields in the same order. Users compare projects, then find the same information in the same place each time. Inconsistent listings would bring back the fragmented presentation that the platform replaces.

7.1.1. NFR-3, Language Support

The platform is viewable in both English and German. German-language requirements were named as a disadvantage by international students (S5, S7, S10), particularly at institutions with a large international intake and degree programs taught in English.
Source: S5, S7, S10

7.2. Security and Data Privacy

7.2.1. NFR-1, Identity Verification

Use of institutional logins for verifying student and staff identities. A verified company email domain serves as a simple check for companies, before being made available to administrative staff for a more in-depth check.
Source: C4

7.2.2. NFR-2, Data Protection for Submitted Documents

CVs, transcripts, and motivation letters uploaded with an application (FR-11) count as personal data under the GDPR. Access is limited to the professor and the company that review the application. No other user can open these files or reach them through project listings. Student-submitted topics (FR-8) and the entries a company accepts (FR-4) stay visible to the other platform roles, but the documents attached to an application do not. How long these documents are stored, and when they are deleted once an application closes, lies outside the scope of this specification and the prototype (Section 4.6).
Source: Process Documentation (legal/regulatory constraint)

7.3. Maintainability / Extensibility

The requirements are structured so that Could-priority and out-of-scope features can be added later without changing the core Must-priority workflow. Student Team Matching (FR-10) and Lightweight Progress Check-in (FR-14) are extensions to the platform’s functionality without requiring changes to it.

8. Use Case Diagrams

The use case diagrams show how the requirements from Section 6 and Section 7 are realised through interactions between students, companies, professors, and university staff. They group the requirements into four connected areas of the platform: access, project intake, application and selection, and profiles and topics. For spacing reasons, the Figures 8.1 - 8.4 of the use case diagrams are placed together after the their description.

8.1. Platform Access

Figure 8.1 shows the common access flow. Users log in and are directed to role-specific dashboards (FR-1). Affiliation is verified during this process (NFR-1), while NFR-3 allows the platform to be viewed in German or English.

8.2. Project Intake

Figure 8.2 shows how projects enter the platform. Companies submit proposals (FR-2), which staff review and approve (FR-3), supported by filtering and status monitoring (FR-17, FR-18). Professors can view relevant submissions, take on supervision, and publish projects (FR-5, FR-6). Directly agreed projects can instead be submitted through FR-7.

8.3. Profiles and Topics

Figure 8.3 covers student profiles and student-initiated topics. Students can submit topics (FR-8), maintain profiles (FR-9), and indicate that they are seeking teammates (FR-10). Companies can browse and accept student-submitted topics through FR-4, completing the required company and contact information when accepting a topic.

8.4. Application and Selection

Figure 8.4 covers the process after publication. Students browse projects, apply, and track their status (FR-11, FR-12). Professors and companies participate in the selection decision (FR-15), while application documents remain access-restricted (NFR-2). Status changes trigger notifications (FR-13), and students can confirm one offer while withdrawing others (FR-16). Progress check-ins are covered by FR-14.

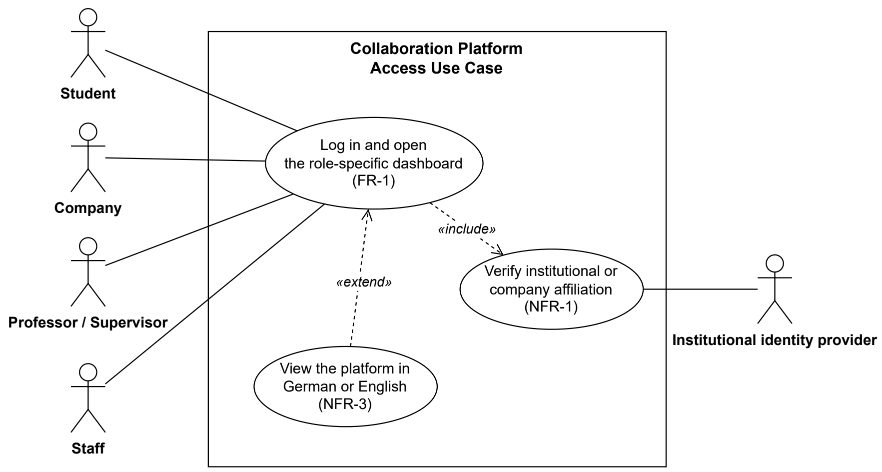

Figure 8.1: Platform access Use Case diagram

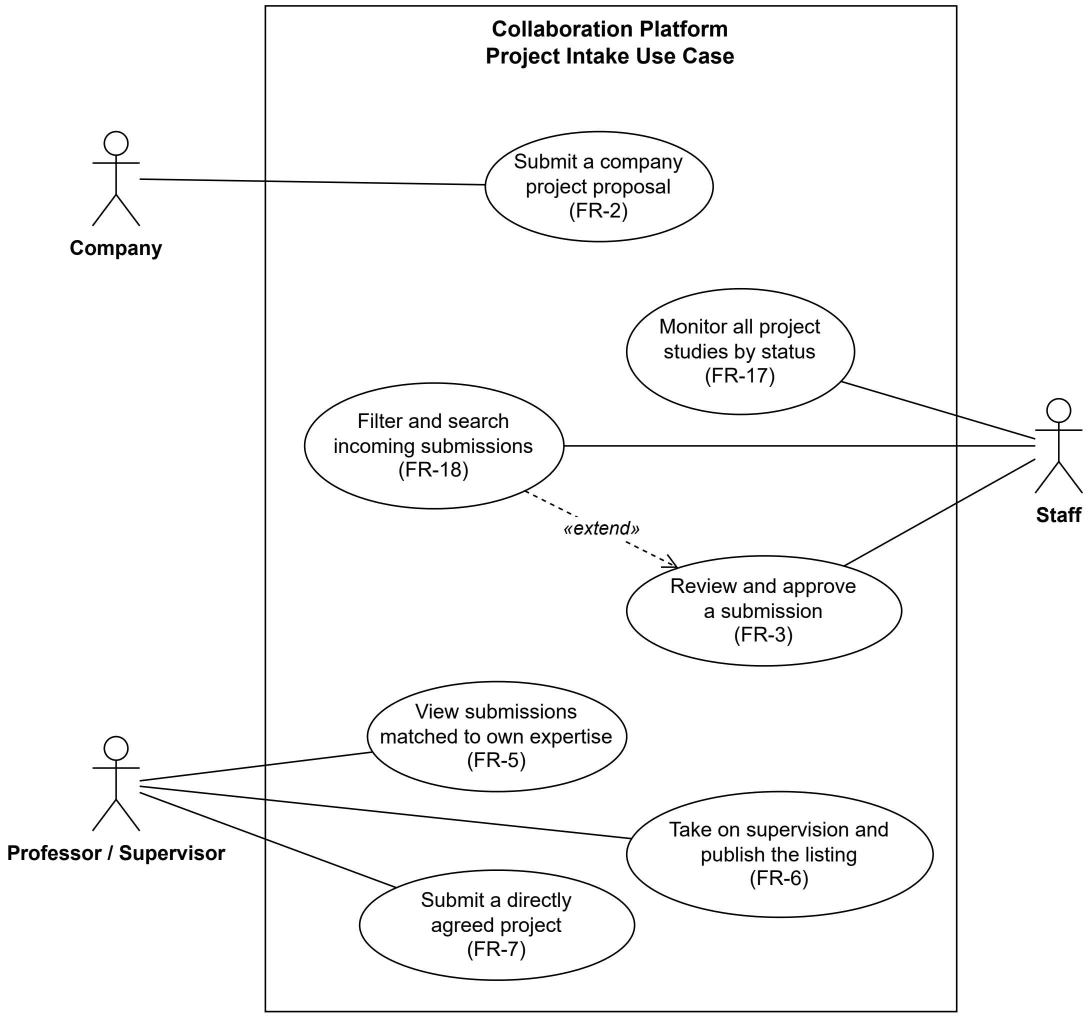

Figure 8.2: Project Intake Use Case diagram

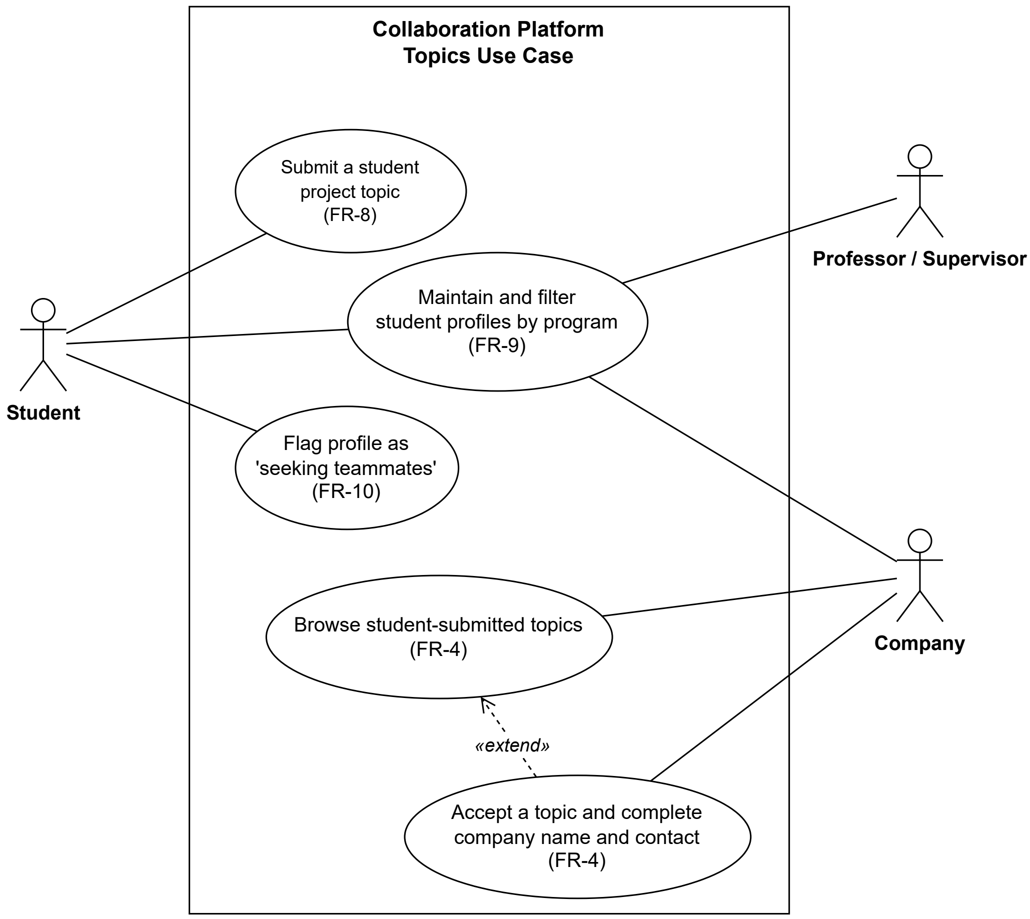

Figure 8.3: Profiles and Topics Use Case diagram

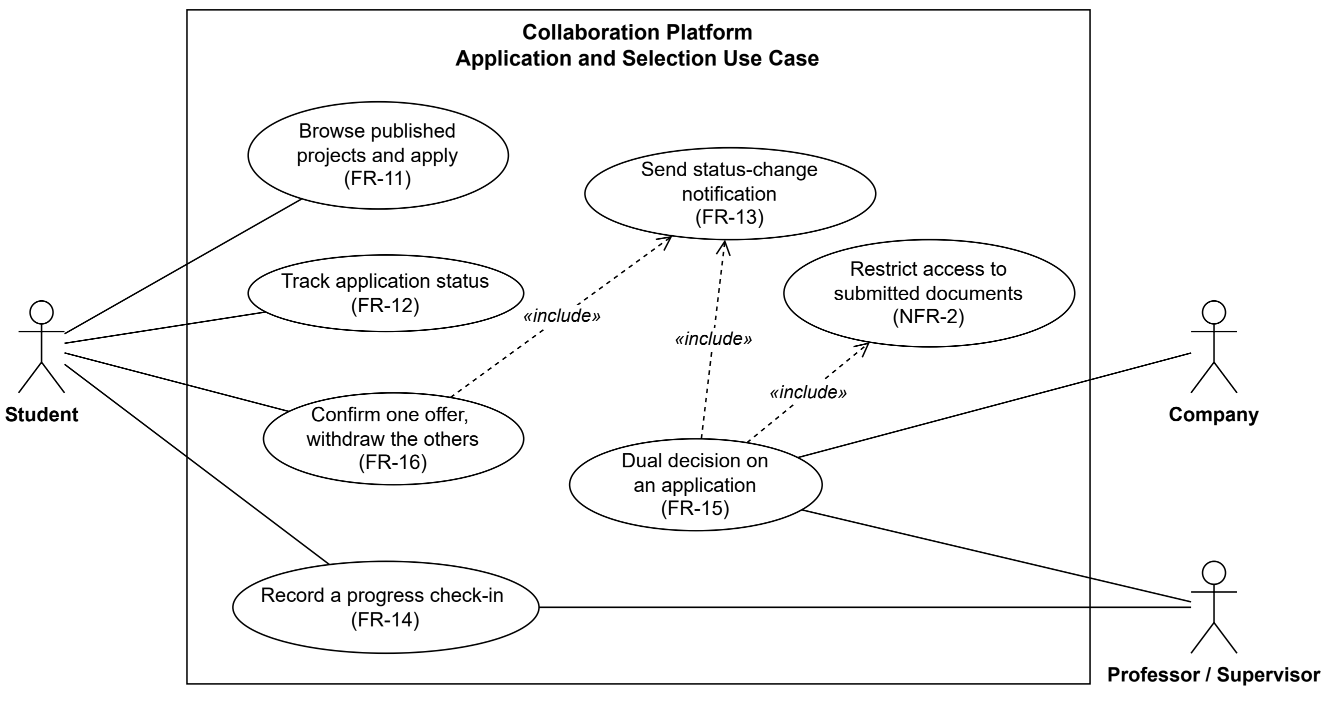

Figure 8.4: Application and Selection Use Case diagram

9. Traceability and Prototype Alignment

9.1. Traceability Table

The full traceability table, including prototype status, MoSCoW priority, and transferability tags for every requirement, is provided in Table A.3. The table below presents the abbreviated form for reference within the running text.

Figure 9.1: Requirements Traceability Overview

Req. ID	Pain Point Summary/Basis	Source Interview
	Access Features	
FR-1	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation
NFR-1	Need for trust and verification	C4
	Company Features	
FR-2	Missing or insufficient listing information, information standardization	S7, S8, S9, P3, P4, P6
FR-3	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation
FR-4	Absence of a clear overview of available posting/ collaboration options; difficulty finding a supervisor	C2 (directly suggested this feature), C5, S12
	Professor / Supervisor Features	
FR-5	Contact and coordination burden	P3, P4, U1, U2
FR-6	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation
FR-7	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation
	Student Features	
FR-8	Absence of a clear overview of available posting/ collaboration options; application process friction	C2, C4, C5, S12
FR-9	Enables student team matching; allows students to be filtered by degree by professors / supervisors and companies	Team Decision
FR-10	Team Formation	S10
FR-11	Discoverability; missing or insufficient listing information	S6, S7, S8, S9
FR-12	Application process friction; communication and responsiveness	S1, S8, S12, S13
	General Features	

Continued on next page

NFR-2	Personal data privacy	Process Documentation
NFR-3	Many international students	S5, S7, S10
FR-13	Contact and coordination burden; communication and responsiveness	S8, U1, U2
FR-14	Students' engagement with professor / supervisors during the project	P4
FR-15	More say in student selection can increase company satisfaction with project studies	Team Decision
FR-16	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation
Staff Features		
FR-17	Contact and coordination burden	U1, U2
FR-18	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation

9.2. Prototype Coverage Table

Every requirement in Section 6 and Section 7 is built in the prototype except for two, both limited by the prototype's frontend-only scope (Section 4.6) rather than by an implementation gap within a single requirement.

- **FR-4 Company Browse:** Companies can browse student-submitted topics and the accept action exists, but clicking it does not yet open the prompt for entering the mandatory acceptance fields (company name, contact information). Since company profiles are not required elsewhere in the prototype, they have to enter this information manually.
- **NFR-2 Data Protection for Submitted Documents:** Requires a backend hosted in a server with persistent storage, which is outside the platform prototype scope.

9.3. Pain Points Without Requirements and Justifications

The following pain points were identified but deliberately left without a corresponding requirement.

- **Language barrier** (S5, S7, S10): Partially addressed. Language Support (NFR-3) makes the platform itself usable in German and English, but the German-language requirements of the projects and chairs themselves lie outside what a platform can change.
- **Value and goal misalignment** (S6, S8, S11, U1, C2, C4): Dual Approval (FR-15) addresses company satisfaction with the selection, but not the underlying motivational mismatch, which cannot be addressed via a platform.
- **Outcome measurement** (P1, P5, C2): Defining an accurate quality measurement is highly complex.
- **Time and financial resource pressure** (P5): External constraint, not addressable via a platform.

- **University structural complexity and inconsistency across institutions** (C4): Acknowledged in the transferability discussion; not resolvable by a feature.
- **Output quality and software/tooling mismatches between student and company environments** (C2): Structural, and therefore largely not addressable via a platform. A further implementation of the light progress check (FR-14) could improve outcome quality by supporting more effective communication.
- **Screening signal degradation due to AI-generated CVs** (C3): Platform-side AI checks are technically conceivable, but their accuracy is low and the implementation effort is not proportionate to the value.

10. Risks, Discussion and Conclusion

This document has translated a set of semi-structured interviews across stakeholder groups into scoped requirements, which can be traced back to the corresponding pain point and traced forward to the prototype coverage, for a project studies platform specific to TUM Campus Heilbronn. This traceability, rather than the completeness of any one requirement, is the main assertion of this document.

Risks primarily rest on 26 interviews of uneven depth across groups, and especially since requirements were not checked back with the people they originated from. So the contents of this document are informed proposals, not validated findings. Two points follow from that directly. Company and professor findings come from smaller, less saturated samples than the student group, so they're less likely to hold up even within TUM Campus Heilbronn specifically. And Dual Approval (FR-15) is a proposed change, not something already happening. The reasoning behind it (companies having more of a say in the process improves satisfaction with outcomes) holds up logically, but nothing actually tests whether it would be effective or introduce further complications. One more risk is not related to the interviews at all: the platform handles personal application data (Section 7.2), and no one with legal expertise has reviewed what that would require.

The transferability tags in the traceability table give pointers on how this work can be extended beyond TUM Campus Heilbronn, but that's discussed in principle in Section 4.5 and nothing more, it hasn't been tried. Closer to what's actually in hand are the broader industry-academia findings collected: experiences, motivations, barriers, what keeps people from collaborating in the first place.

References

Achimugu, P., Selamat, A., Ibrahim, R., & Mahrin, M. (2014). A systematic literature review of software requirements prioritization research. *Information and Software Technology*, *56*, . https://doi.org/10.1016/j.infsof.2014.02.001

Ankrah, S., & AL-Tabbaa, O. (2015). Universities-industry collaboration: A systematic review. *Scandinavian Journal of Management*, *31*(3), 387–408. https://doi.org/https://doi.org/10.1016/j.scaman.2015.02.003

Bruneel, J., D'Este, P., & Salter, A. (2010). Investigating the factors that diminish the barriers to university-industry collaboration. *Research Policy*, *39*(7), 858–868. https://doi.org/https://doi.org/10.1016/j.respol.2010.03.006

Ferrari, A., Spoletini, P., & Debnath, S. (2022,). *How Do Requirements Evolve During Elicitation? An Empirical Study Combining Interviews and App Store Analysis*. https://arxiv.org/abs/2208.00825

Gotel, O., & Finkelstein, A. (1994). An analysis of the requirements traceability problem. *Proceedings of IEEE International Conference on Requirements Engineering*, 94–101. https://api.semanticscholar.org/CorpusID:5870868

Guest, G., Bunce, A., & Johnson, L. (2006). How Many Interviews Are Enough?: An Experiment with Data Saturation and Variability. *Field Methods*, *18*, 59–82. https://doi.org/10.1177/1525822X05279903

Hagaman, A. K., & Wutich, A. (2017). How Many Interviews Are Enough to Identify Metathemes in Multisited and Cross-cultural Research? Another Perspective on Guest, Bunce, and Johnson's (2006) Landmark Study. *Field Methods*, *29*(1), 23–41. https://doi.org/10.1177/1525822X16640447

Hennink, M. M., Kaiser, B. N., & Marconi, V. C. (2017). Code Saturation Versus Meaning Saturation: How Many Interviews Are Enough?. *Qualitative Health Research*, *27*(4), 591–608. https://doi.org/10.1177/1049732316665344

Howells, J. (2006). Intermediation and the role of intermediaries in innovation. *Research Policy*, *35*(5), 715–728. https://doi.org/https://doi.org/10.1016/j.respol.2006.03.005

MacQueen, K. M., McLellan, E., Kay, K., & Milstein, B. (1998). Codebook Development for Team-Based Qualitative Analysis. *CAM Journal*, *10*(2), 31–36. https://doi.org/10.1177/1525822X980100020301

Perkmann, M., Tartari, V., McKelvey, M., Autio, E., Broström, A., D'Este, P., Fini, R., Geuna, A., Grimaldi, R., Hughes, A., Krabel, S., Kitson, M., Llerena, P., Lissoni, F., Salter, A., & Sobrero, M. (2013). Academic engagement and commercialisation: A review of the literature on university-industry relations. *Research Policy*, *42*(2), 423–442. https://doi.org/https://doi.org/10.1016/j.respol.2012.09.007

Society, I. C. (2018). *ISO/IEC/IEEE 29148:2018; ISO/IEC/IEEE International Standard - Systems and software engineering – Life cycle processes – Requirements engineering*. https://standards.ieee.org/ieee/29148/6937/

Appendices

Table A.1: Sample Composition, Participant IDs, and Data Collection Format ($n = 26$)

ID	Interviewee/Response	Stakeholder Group	Format
S1	TUM Campus Garching Student 1	Student	Interview
S2	TUM Campus Heilbronn Student 1	Student	Interview
S3	TUM Campus Garching Student 2	Student	Interview
S4	TUM Campus Heilbronn Student 2	Student	Interview
S5	TUM Campus Heilbronn Student 3	Student	Interview
S6	RWTH Aachen University Student	Student	Written response
S7	TUM Campus Heilbronn Student 4	Student	Written response
S8	TUM Campus Heilbronn Student 5	Student	Interview
S9	TUM Campus Heilbronn Student 6	Student	Interview
S10	TUM Campus Heilbronn Student 7	Student	Written response
S11	Reutlingen University Student	Student	Written response
S12	TUM Campus Heilbronn Student 8	Student	Interview
S13	TUM Campus Heilbronn Student 9	Student	Interview
C1	Founder of IT Solution Startup	Company	Interview
C2	IDS HR Business Partner	Company	Interview
C3	Atlas Copco Manager Apprenticeship	Company	Written response
C4	SAP Campus Ambassadors Lead	Company	Written response
C5	Schneider HR Department	Company	Written response
P1	Koblenz University Professor	Professor (external)	Written response
P2	TUM Campus Munich Professor	Professor (external, other campus)	Written response
P3	TUM Campus Heilbronn Professor	Professor (TUM Heilbronn)	Interview
P4	TUM Campus Heilbronn Professor	Professor (TUM Heilbronn)	Interview
P5	RWTH Aachen University Professor	Professor (external)	Written response
P6	Passau University PhD Candidate	Research Candidate (external)	Interview
U1	TUM Campus Heilbronn Administrative Staff 1	University staff (TUM Heilbronn)	Interview
U2	TUM Campus Heilbronn Administrative Staff 2	University staff (TUM Heilbronn)	Interview

Table A.2: Thematic Areas and Guiding Questions

#	Topic	Guiding Question
1	Current collaboration experience	Tell us about your experience with academia-industry collaboration (e.g. how you found or engaged in it, application/agreement process, communication with the other side).
2	Motivation	What motivates you to take part in this kind of collaboration?
3	Problems and pain points	What problems or pain points have held you back, or made this harder than it needed to be?
4	Platform/solution needs	If there were a platform supporting this kind of collaboration, what would you want to see or find useful?

Table A.3: Full Requirements Traceability Matrix

Req. ID	Function/ Feature	Description	Pain Point Summary/ Basis	Source Interview	MoSCoW	Transferability	Prototype Status
Access Features							
FR-1	Role-Based Access (4 roles)	Four distinct login/user types (Student, Company, Professor / Supervisor, Staff). Each must have a tailored dashboard	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation	Must	Adaptable	Built
NFR-1	Identity Verification	Reuse institutional logins for student/staff identity. Verified company email domain as a lightweight company check	Need for trust and verification	C4	Must	Generic	Built
Company Features							
FR-2	Company Project Submission Portal	Mandatory fields include: Project Title, Required Area of expertise, Project Background and Objective, Project Deliverable, Available Company Resources, Required Student Skills, Required Group Size, Company Contact Person (adapted from the current project proposal sheet)	Missing or insufficient listing information, information standardization	S7, S8, S9, P3, P4, P6	Must	Adaptable	Built
FR-3	Submission Review and Approval	New project submissions recorded as 'pending'; staff approve before professor and supervisor visibility	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation	Should	TUM-specific	Built

Req. ID	Function/ Feature	Description	Pain Point Summary/ Basis	Source Interview	MoSCoW	Transferability	Prototype Status
FR-4	Company Browse	Companies browse student-submitted topics; acceptance requires the company to fill in mandatory fields (company name, company contact information), and the completed entry becomes visible to professors and supervisors	Absence of a clear overview of available posting/collaboration options; difficulty finding a supervisor	C2 (directly suggested this feature), C5, S12	Must	Generic	Partial
Professor / Supervisor Features							
FR-5	Professor / Supervisor Profiles and Expertise Matching	Professor / supervisor profiles list chair/expertise (can be sourced from institution login); on login, professors / supervisors see project entries matched to their area of expertise	Contact and coordination burden	P3, P4, U1, U2	Should	Generic	Built
FR-6	Supervision Take-on	To accept a project submission, professor / supervisor fills in rest of the required information (chair contact information, application deadline and list required application documents); project becomes visible for students	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation	Must	Adaptable	Built

Req. ID	Function/ Feature	Description	Pain Point Summary/ Basis	Source Interview	MoSCoW	Transferability	Prototype Status
FR-7	Direct Project Submission (Professor / Supervisor + Company Agreement)	When a company and professor agree directly to offer a project study, professors / supervisors can submit it with the mandatory fields filled directly for the students to apply	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation	Should	Adaptable	Built
			Student Features				
FR-8	Student Project Submission Portal	Mandatory fields include: Project Title, Required Area of expertise, Project Background and Objective, Project Deliverable, Required Company Resources, Group Size, Student Team Contact Person (adapted from the current project proposal sheet)	Absence of a clear overview of available posting/collaboration options; application process friction	C2, C4, C5, S12	Must	Adaptable	Built
FR-9	Student Profiles	Student profiles list program/ degree (can be sourced from institution login)	Enables student team matching; allows students to be filtered by degree by professors / supervisors and companies	Team Decision	Should	Generic	Built

Req. ID	Function/ Feature	Description	Pain Point Summary/ Basis	Source Interview	MoSCoW	Transferability	Prototype Status
FR-10	Student Team Matching	Student chooses to set the status on their profile to 'seeking teammates' + field for a small message and contact information, profiles are filterable by this status	Team Formation	S10	Could	Generic	Built
FR-11	Student Browse Available Projects and Apply	Students view available projects with full info (company, expertise, application deadline, skills) and contact person; they apply for a given entry by submitting the required documents	Discoverability; missing or insufficient listing information	S6, S7, S8, S9	Must	Generic	Built
FR-12	Application Status Visibility	Simple accepted/rejected status per application	Application process friction; communication and responsiveness	S1, S8, S12, S13	Should	Generic	Built
General Features							
NFR-2	Data Protection for Submitted Documents	Only the specific professor / supervisor and company reviewing an application can see the application materials submitted	Personal data privacy	Process Documentation	Must	Generic	Not Planned
NFR-3	Language Support	The platform should be viewable in German and English.	Many international students	S5, S7, S10	Could	Generic	Built

Req. ID	Function/ Feature	Description	Pain Point Summary/ Basis	Source Interview	MoSCoW	Transferability	Prototype Status
FR-13	Status-Change Notifications	Automated notification (e.g. email) when an application's status changes or a listing closes	Contact and coordination burden; communication and responsiveness	S8, U1, U2	Should	Generic	Built
FR-14	Lightweight Progress Check-in	Simple, non-rigid check-in mechanism between student and professor / supervisor during the project	Students' engagement with professor / supervisors during the project	P4	Could	Adaptable	Built
FR-15	Dual Approval (Professor/ Supervisor + Company)	A student application requires both professor / supervisor and company sign-off before being finalized	More say in student selection can increase company satisfaction with project studies	Team Decision	Could	Adaptable	Built
FR-16	Manual Multi-Offer Decision	If accepted to multiple projects, student manually confirms one and manually withdraws the others	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation	Should	Generic	Built
Staff Features							
FR-17	Staff Master Dashboard	View all project studies across every status: pending, approved, not yet supervised, ongoing, filled	Contact and coordination burden	U1, U2	Should	TUM-specific	Built
FR-18	Filter Incoming Submissions	Staff can filter/search incoming company submissions during review	Adapted from the current process documentation in TUM Campus Heilbronn	Process Documentation	Should	TUM-specific	Built